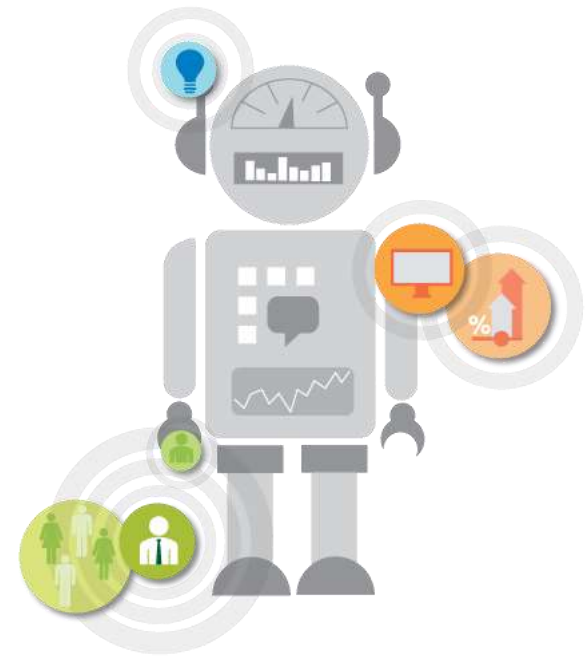


AI and Society

CSET-485

Dr. Susmita Das



UNIT - 2



What is Feature Importance?

Feature Importance

Feature importance is a technique used to determine **how much each input variable contributes to a machine learning model's predictions.**

Why Feature Importance is Needed?

In complex models (Random Forest, Gradient Boosting, Neural Networks), we often ask:

- Which features influence predictions the most?
- Are irrelevant features affecting the model? Criminal justice (risk assessments)
- Is the model relying on sensitive attributes (e.g., gender, caste, age)?

What Feature Importance Tells

- Ranking of features
- Which variables drive predictions
- Helps in feature selection
- Identifies bias or leakage

Types of Feature Importance

1. Model Specific Feature Importance

Some models naturally compute importance.

Example: In tree-based models, importance is calculated based on **impurity reduction**. In Gini Index for decision tree, we calculate the reduction in impurity.

$$Gini = 1 - \sum_{i=1}^n (p_i)^2$$

When Gini Index is small, it means less impurity, it can be considered as important feature for classification.

Types of Feature Importance

2. Permutation Feature Importance (Model-Agnostic)

This method works with any model. Here:

- Train model normally
- Shuffle values of one feature
- Measure drop in performance

$$Importance = Performance_{original} - Performance_{shuffled}$$

If model performance drops significantly → feature is important.

3. SHAP Based Feature Importance

Using SHAP we compute

$$Global\ Importance = \text{mean}(|SHAP\ value|)$$

This measures the average contribution magnitude of each feature.

Model Interpretability

Model interpretability in Explainable AI (XAI) focuses on **understanding how and why a model makes decisions**. It is broadly divided into:

- **Global Interpretation** → Understanding the model's overall behavior
- **Local Interpretation** → Understanding a single prediction

Global Interpretation: Global interpretation explains **how the entire model works across the whole dataset**.

It explains the questions, the features that are generally important, the patterns the model has learned and the overall decision logic

Local Interpretation: Local interpretation explains **why the model made a specific prediction for one instance**.

It basically explains the individual decisions related to each feature.

Global Model Interpretability

Intrinsically Interpretable Models

These are models that are naturally explainable. Some models with such weights are Linear Regression, Logistic Regression, Decision Trees, Random Forest, Rule-based Models.

Example: For Linear Regression, coefficients β_1, β_2 directly show global feature importance.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Feature Importance (Model-Agnostic)

Measures how much each feature contributes overall.

Global SHAP Values

Using average magnitude of local SHAP values

$$Global\ Importance(j) = \frac{1}{n} \sum_{i=1}^n |\phi_j(x^{(i)})|$$

Partial Dependence Plots (PDP)

Shows how a feature affects predictions on average.

Partial Dependence Plots

Partial Dependence Plots (PDP)

A Partial Dependence Plot (PDP) is a global interpretation technique that shows, **the effect of a selected feature on the model's prediction on average**, while averaging out the effects of all other features.

Example: This helps in determining, whether income increase loan approval probability or whether age linearly related to risk in loan approval,

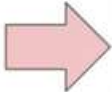
Need for PDP

In complex models like Random Forest, Gradient Boosting, Neural Networks etc, relationship between input and output is not directly visible.

PDP helps visualize **a) Direction of relationship (positive/negative)**, **b) Non-linearity in the relationship**, **c) Threshold effects**

Partial Dependence Plots

Let us consider a dataset with details on the **sales of 1000 second-hand cars**. The dataset consists of **5 features** as listed below. Goal is to **predict the price** of the second-hand car based on the given features.



Name	Description
owner_age	Age of the previous owner
km_driven	Distance driven at time of sale (kilometers)
car_age	Age of car at time of sale
repairs	The number of times the car has been serviced/repaired
car_type	A normal (0) or classic (1) car
price	The price the second hand car was sold for (\$)

Partial Dependence Plots

Suppose, we already have a model trained on this dataset. Here, we have one instance of this dataset, where the predicted price is given based on the features listed.

owner_age	car_age	km_driven	repairs	car_type	price	Predicted price (yhat)
23	2.17	25,142	24	0	4,077	3524.90

We can change any one of these features and observe the changes in predicted price.

owner_age	car_age	km_driven	repairs	car_type	price	Predicted price (yhat)
23	10	25,142	24	0	4,077	3458.55

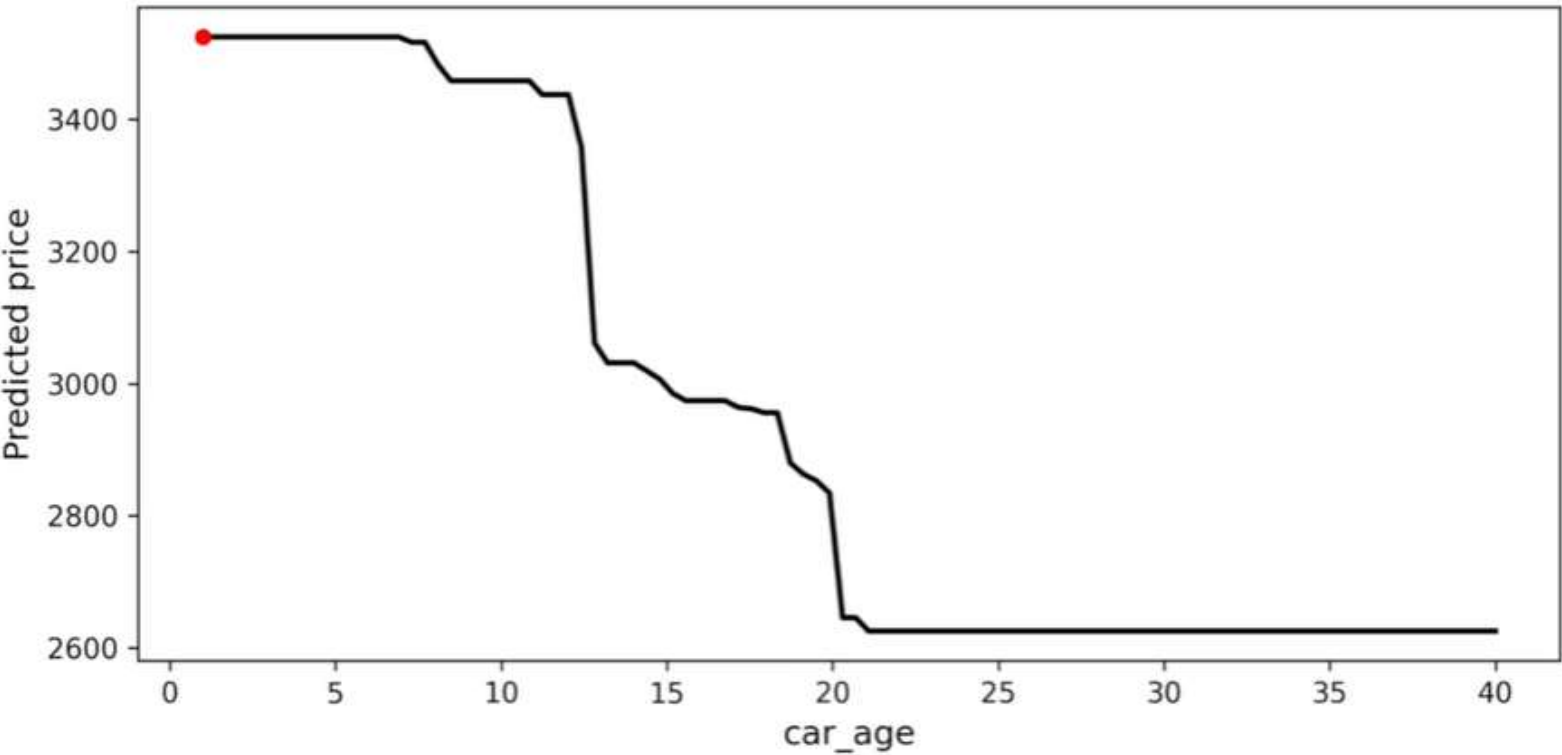
owner_age	car_age	km_driven	repairs	car_type	price	Predicted price (yhat)
23	2.17	25,142	10	0	4,077	5444.24

Partial Dependence Plots

This plot is created by varying the value of car age between its minimum and maximum value.

We hold all the other feature values constant at the original values.

owner_age	car_age	km_driven	repairs	car_type	price	Predicted price (yhat)
23	1	25,142	24	0	4,077	3524.90

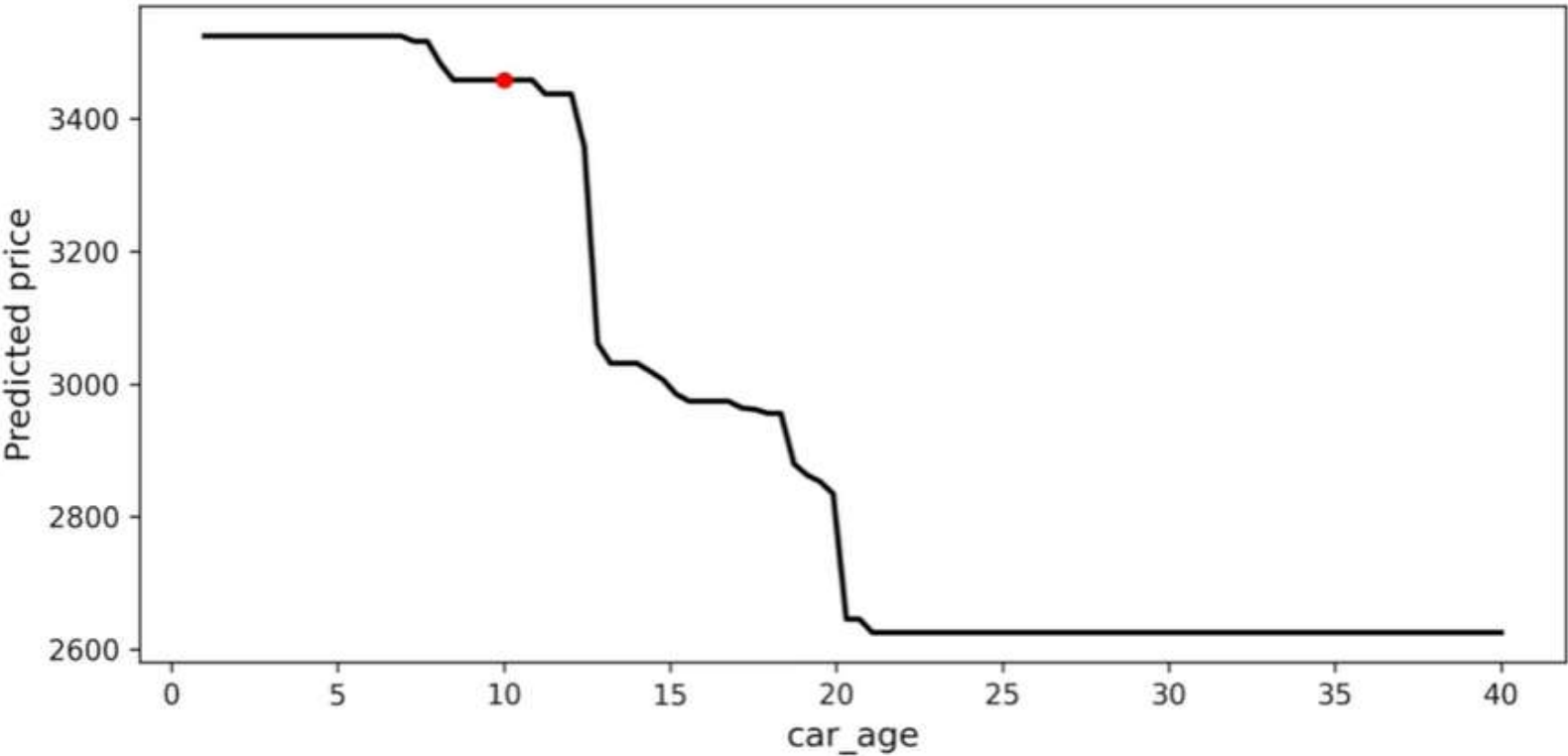


Partial Dependence Plots

This plot is created by varying the value of car age between its minimum and maximum value.

We hold all the other feature values constant at the original values.

owner_age	car_age	km_driven	repairs	car_type	price	Predicted price (yhat)
23	10	25,142	24	0	4,077	3458.56

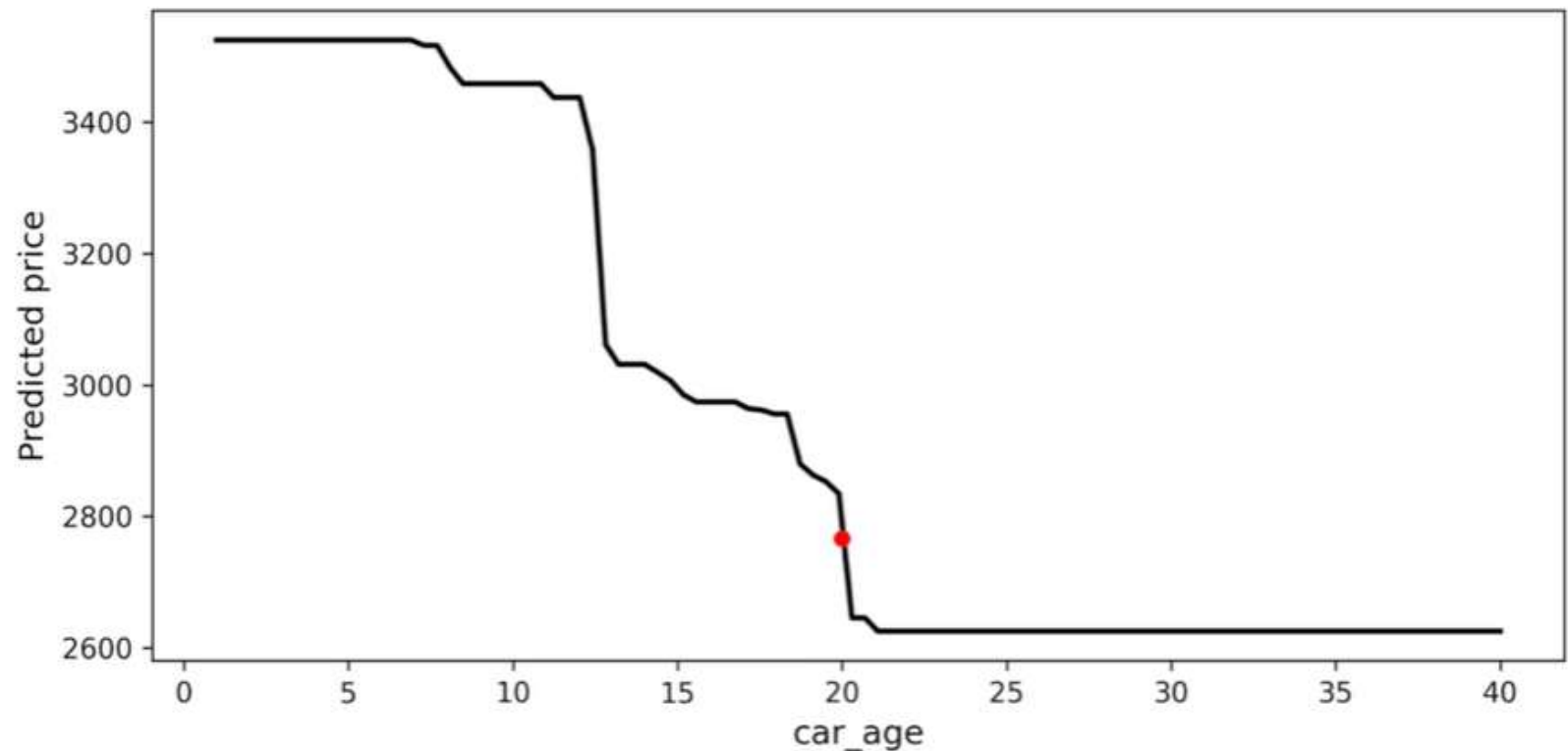


Partial Dependence Plots

This plot is created by varying the value of car age between its minimum and maximum value.

We hold all the other feature values constant at the original values.

owner_age	car_age	km_driven	repairs	car_type	price	Predicted price (yhat)
23	20	25,142	24	0	4,077	2766.94

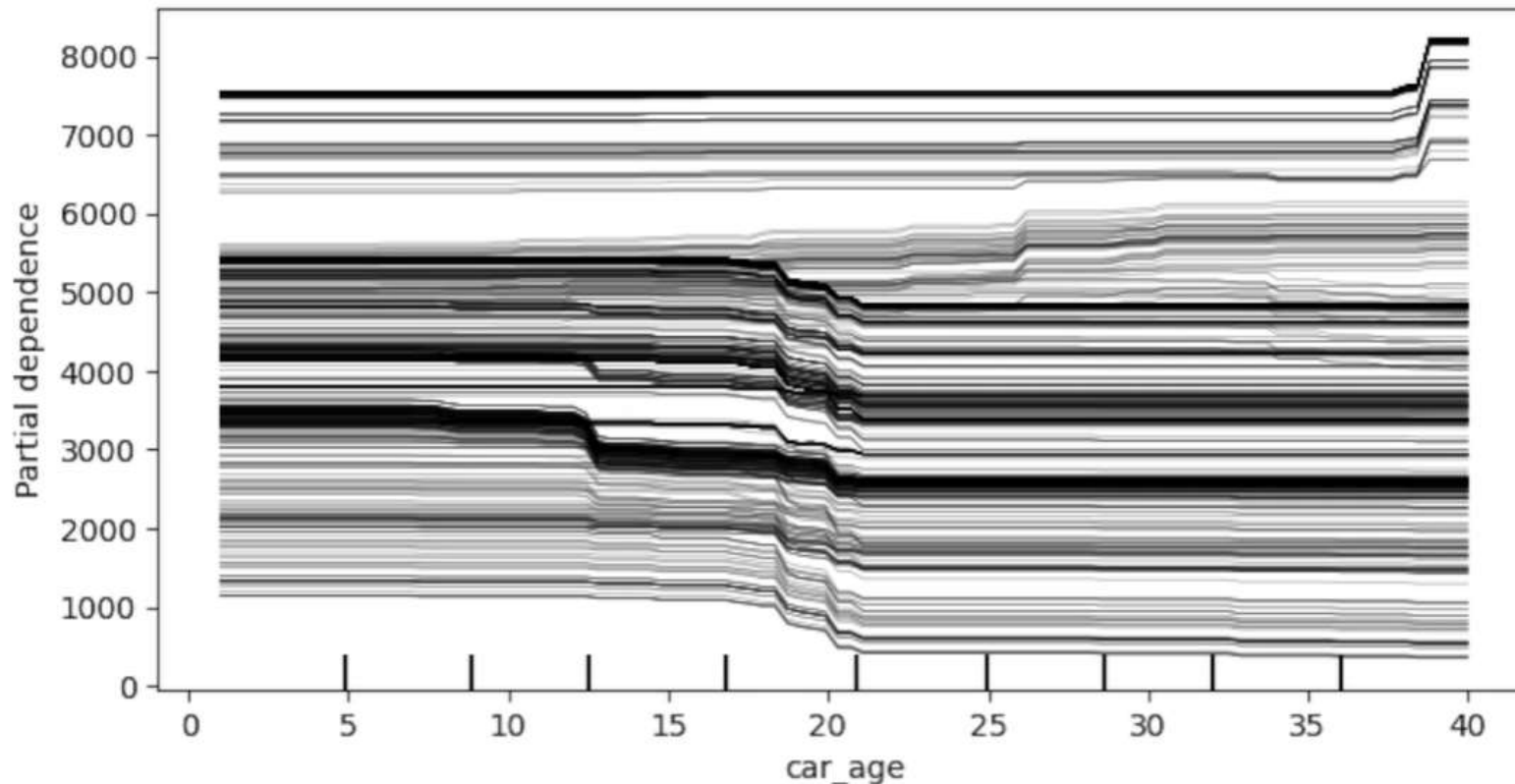


Partial Dependence Plots

Previously, it was the instance of a single data point in the dataset.

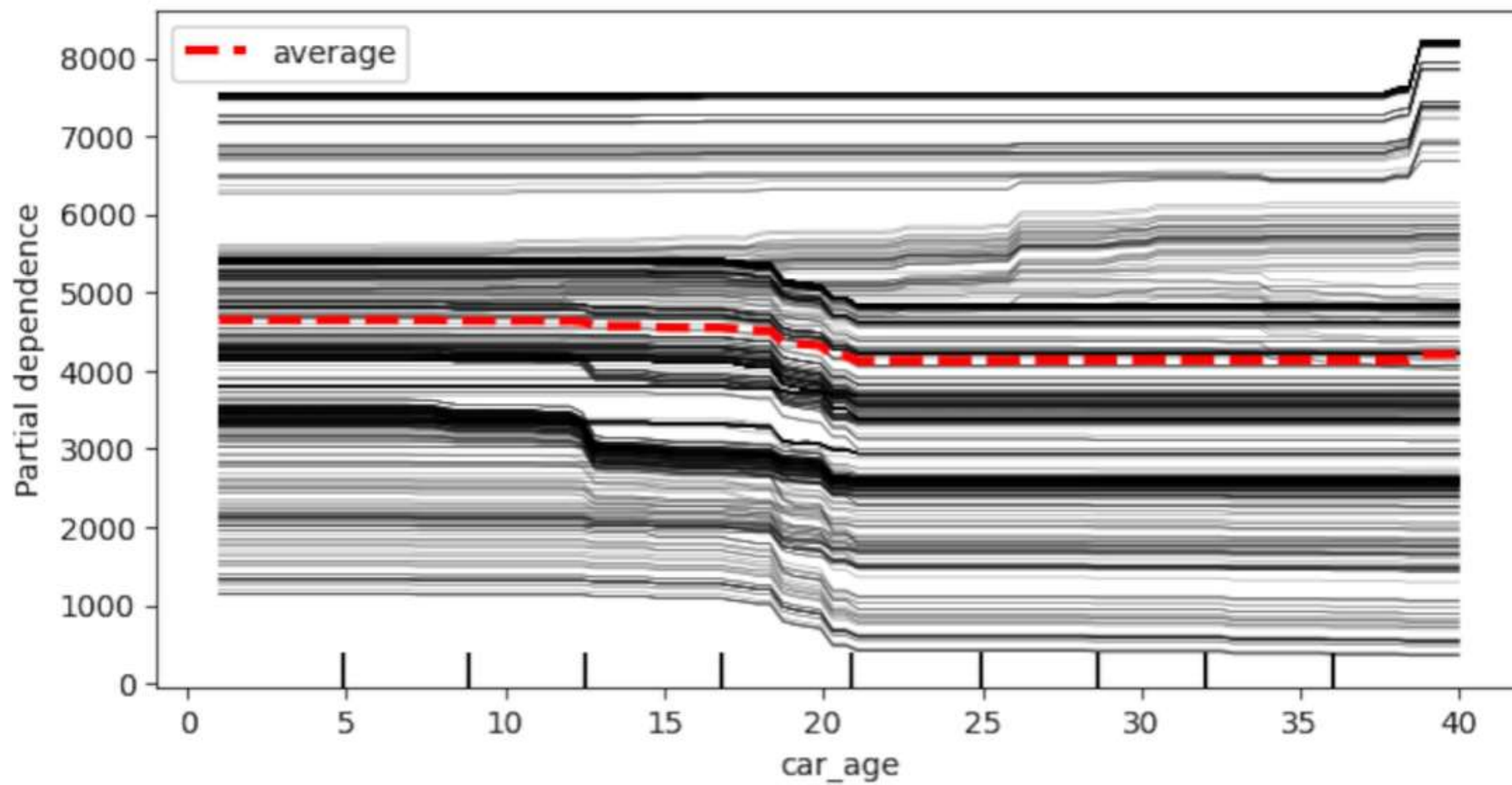
We can repeat this process for every instance in the dataset. In the end, we will obtain 1000 such plots, each corresponding to each instance in the dataset.

Note: For each instance, we have only one feature varied i.e. car age



Partial Dependence Plots

To create a PDP, the final step is to calculate the final step to calculate the average prediction at each value for car age. The red line is the PDP.



Partial Dependence Plots

Example: Loan Approval Model

Feature: Income

Output: Approval Probability

For each income level, replace income in dataset and compute average approval probability

Suppose, $\text{Income} < ₹20,000 \rightarrow \text{Low probability}$

$₹20,000 - ₹50,000 \rightarrow \text{Steady increase}$

$₹80,000 \rightarrow \text{Saturation (no major increase)}$

Interpretation: Income positively influences approval, but effect flattens after certain level

Types of PDP

1D PDP: Shows effect of one feature.

2D PDP: Shows interaction between two features.

Local Model Interpretability

SHAP(Local)

SHAP explains **a single prediction** by assigning each feature a contribution value and provides additive feature contributions. For one instance, the formula is:

$$f(x) = \phi_0 + \sum_{j=1}^p \phi_j$$

Saliency Maps (Deep Learning)

A Saliency Map is a visualization technique used in deep learning to show which parts of the input most influenced the model's prediction.

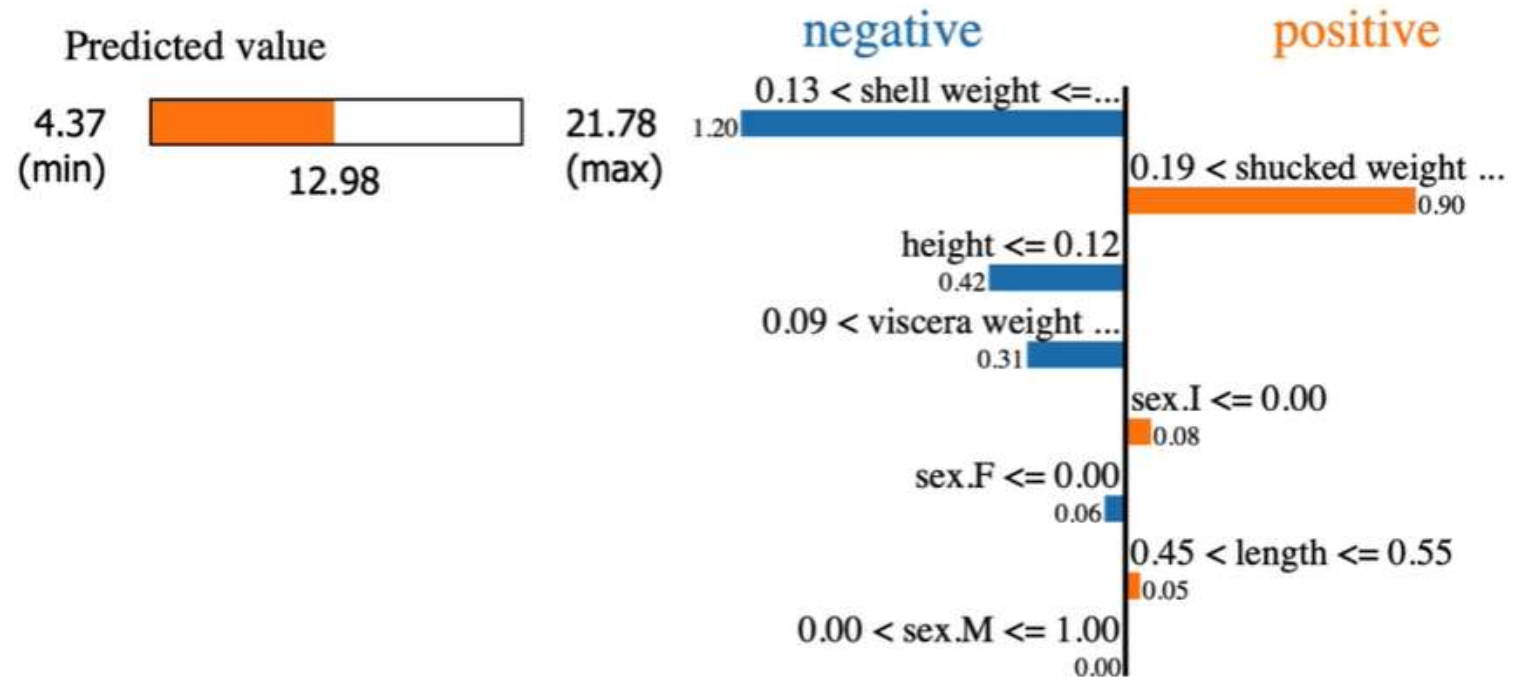
Local Interpretable Model-agnostic Explanations (LIME)

LIME explains **individual predictions** by approximating the complex model locally with a simple interpretable model.

Local Interpretable Model-agnostic Explanations (LIME)

Predict Number of rings in an abalone shell.

The bars provide the line weight. Here, we can see the shell weight and the shucked weight features are the most important. It can also be observed that, they have the opposite effect. We cannot tell exactly how they have contributed to the prediction. The bars just provide relative importance of the features.

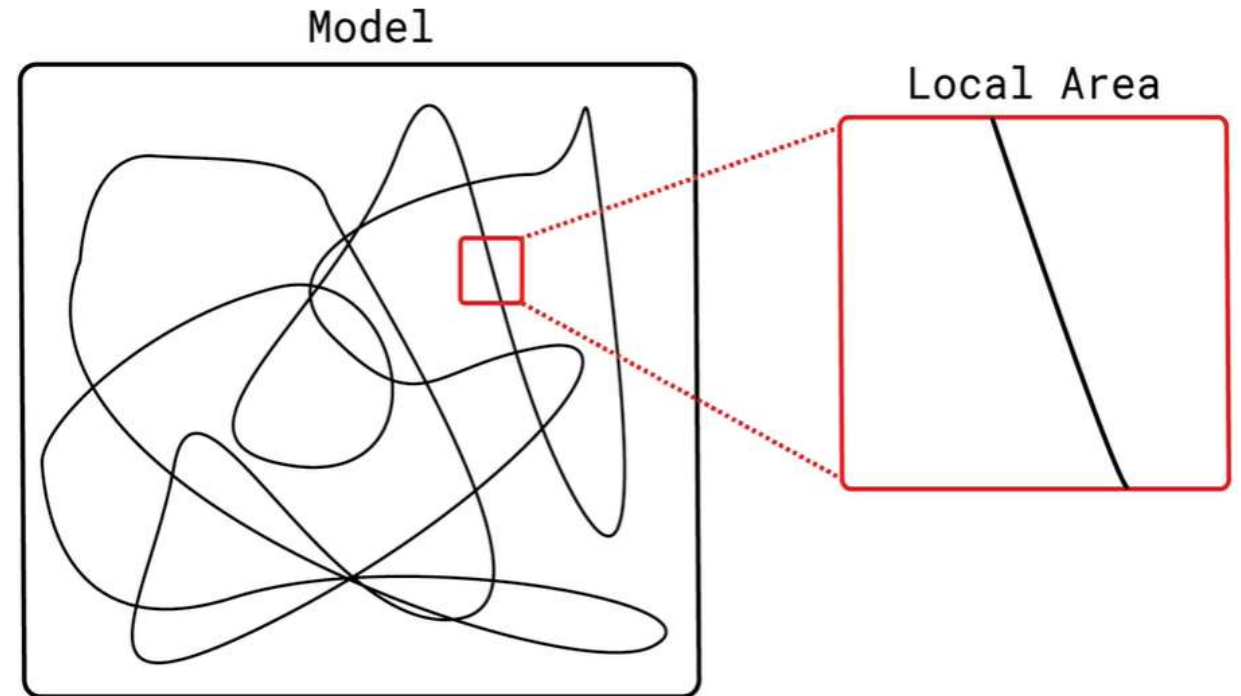


Local Interpretable Model-agnostic Explanations (LIME)

AIML models are complex functions.

LIME says, this complexity is removed, if we zoom into the feature space in the area around an instance.

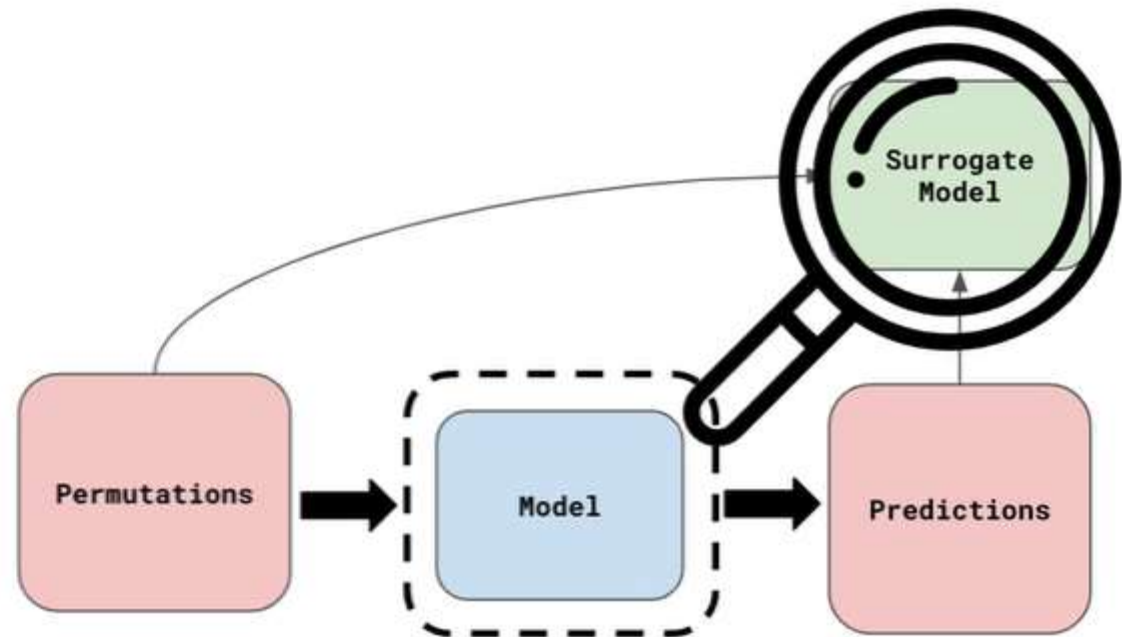
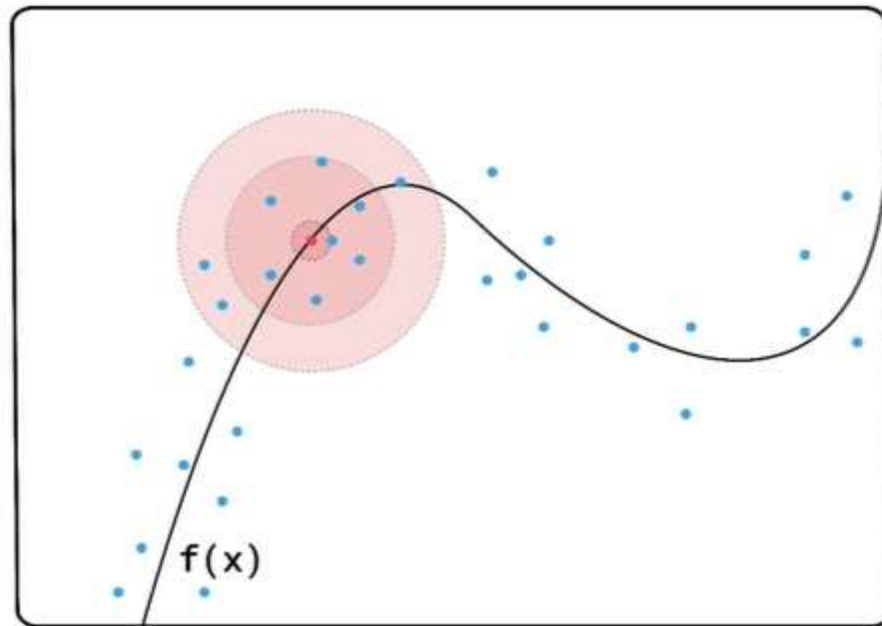
This allows us to understand how predictions are made **using surrogate models**. The surrogate model should be intrinsically interpretable, like linear regression or decision tree.



Surrogate Model is a simple, interpretable model that approximates a complex (black-box) model locally around a specific data instance.

Local Interpretable Model-agnostic Explanations (LIME)

1. Select the instance we want to explain
2. Generate samples
3. Assign a weight to each sample
4. Make predictions on permutations
5. Train a surrogate model
6. Interpret the surrogate model



Local Interpretable Model-agnostic Explanations (LIME)

Limitations of LIME:

For interpretable models, the coefficients for each feature, tell us how the feature has contributed to the prediction. Initially, all of these choices may seem a good thing. But this leads to one of the biggest weakness of LIME. This leads to **inconsistent explanations**.

Designers can **manipulate the variables until they get the explanation they want**. For users, without any domain knowledge, it becomes **difficult to understand whether the explanation is reasonable**.

As a result of this, SHAP has become much more popular than LIME in Explainable AI.